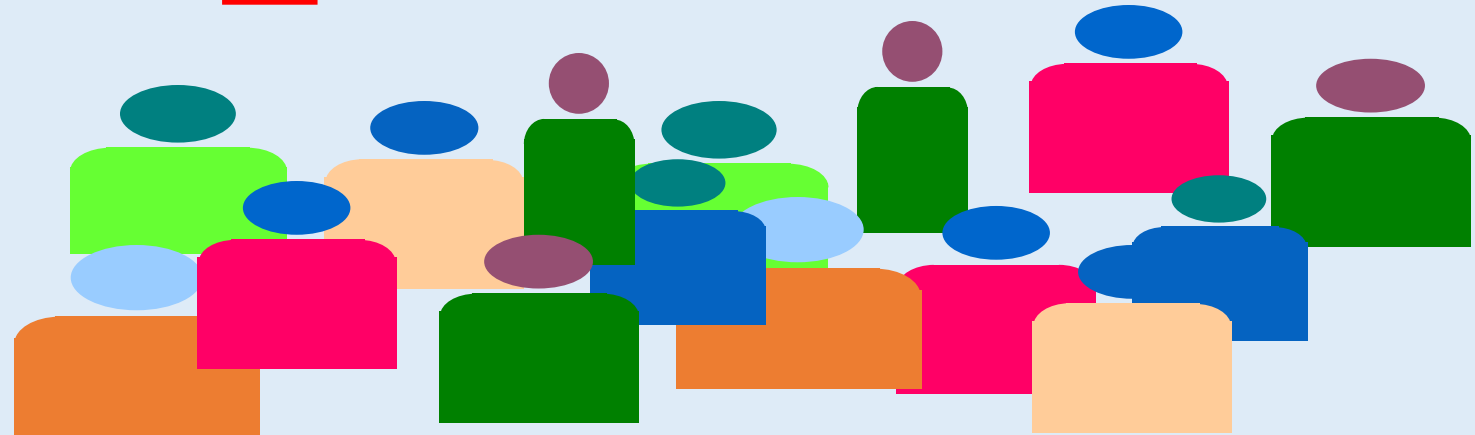
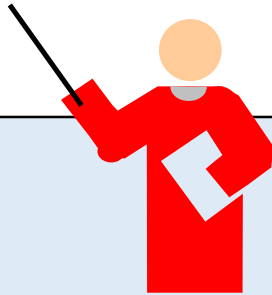
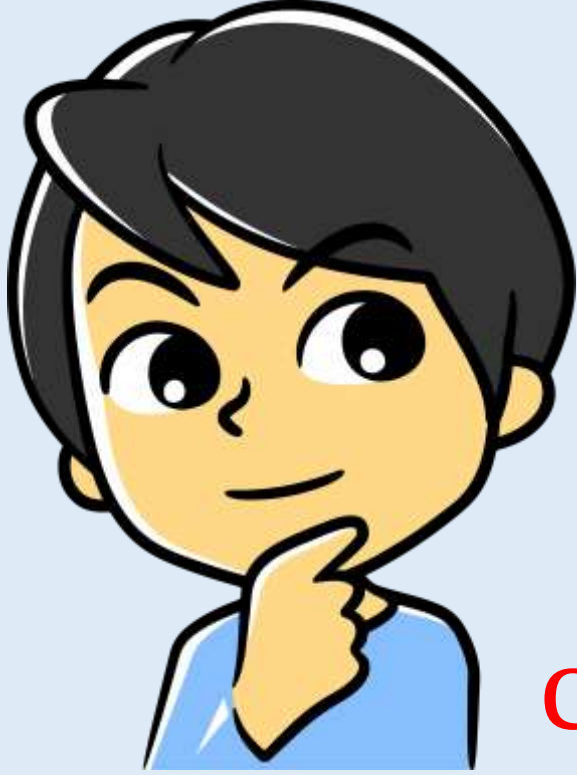


Introduction to Statistics

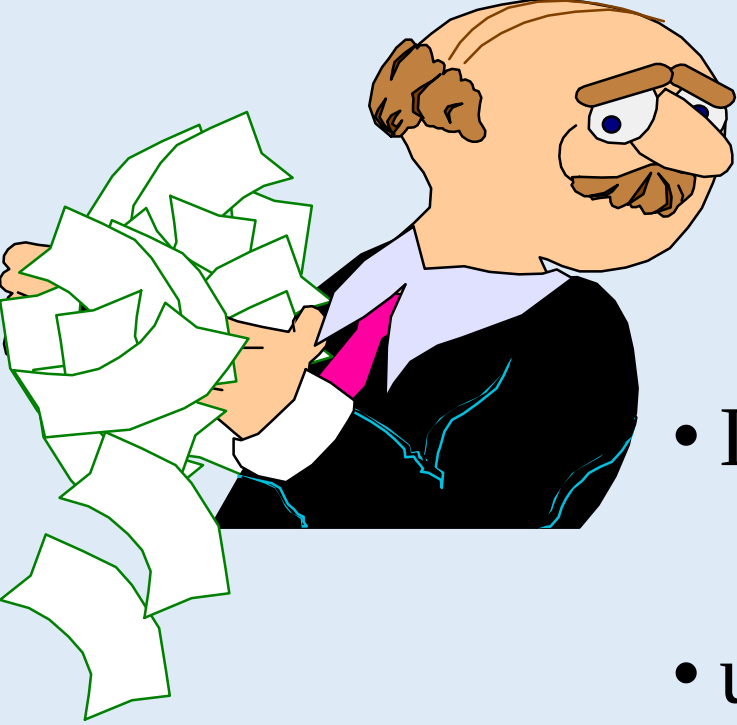
Statistics For AI&DS





What is Statistics?

Statistics is the science of collecting, organizing, analyzing, and interpreting data in order to make decisions.



Origin and Development

- Latin word ‘status’ or ‘statista’
- used to survey the data of population, agriculture, and trade of states, to ensure the manpower and later to regain the taxes.
- during the wars , collecting the information of manpower to meet loss in war and implementation of new takes to balance the loss.

- In ancient **Greece**, scholars like **Plato and Aristotle** made contributions to the field by discussing the use of averages and the importance of data in making decisions.
- **17th Century:** The concept of **probability**, a key component of statistics, began to take shape in the 17th century. **Blaise Pascal and Pierre de Fermat** exchanged letters discussing problems related to games of chance, and their correspondence laid the foundation for probability theory.
- **18th Century:** **Thomas Bayes** introduced Bayesian probability theory, and the work of **Jacob Bernoulli and others** contributed to the understanding of probability and statistical inference.

- **19th Century:** Sir Francis Galton developed concepts like correlation and regression and applied statistical methods to the study of heredity. Karl Pearson, another influential statistician, further developed statistical techniques during this period.
- **20th Century:** Ronald A. Fisher, Jerzy Neyman, and Egon Pearson contributed to the development of hypothesis testing and experimental design. Bayesian statistics also experienced a revival in the mid-20th century.
- **Computational Era:** With the advent of computers in the mid-20th century, statistical analysis became more sophisticated and accessible. The field of statistics expanded to include techniques such as regression analysis, analysis of variance, and non-parametric methods.

History of Statistics

Prehistory of Statistics 1654-1750

Main contributors

- Cardano: 1650
- Pascal y Fermat: 1654
- Graunt: 1662
- Halley: 1693
- Arbuthnot: 1710
- J. Bernoulli: 1713
- De Moivre: 1738
- D. Bernoulli: 1738

The introduction of inference and the beginning of mathematical statistics 1750-1820

Main contributors

- Mayer: 1750
- Simpson: 1755
- Bayes: 1764
- Laplace: 1774
- D. Bernoulli: 1778
- Laplace: 1787
- Legendre: 1805
- Gauss: 1809
- Laplace: 1810

The socialization of statistics and the development of correlation and statistical models 1820-1900

Main contributors

- Quetelet: 1824, 1829, 1935
- Cournot: 1843
- Maxwell: 1859
- Galton: 1869, 1885, 1888
- Edgeworth: 1892
- Yule: 1897
- Pearson: 1900

Modern Statistics foundation 1900-1955

Main contributors

- Fisher: 1915, 1922, 1923, 1925, 1935
- Ramsay: 1926
- De Finetti: 1930
- Kolmogorov: 1933
- Neyman-Pearson: 1933
- Neyman: 1934
- Jeffreys: 1939
- Wald: 1945
- Savage: 1954



- Business and Economics
- Engineering and Technology
- Medical or Biological Fields

Importance of Statistics

- 1) Statistics is important mainly in data analysis.
- 2) It provides a single solution to multiple variational parametric data.
- 3) Statistics sometime used find and establish the relations between physical quantities.
- 4) Statistics also provides the prediction and estimation of certain relevant data.
- 5) It also provides dimensional nature of data like survey, spread of data, divergence of data.
- 6) Statistics support to reach to conclusion of a processed data.
- 7) Statistics itself provides a tool of comparision.
- 8) Uncertainty in data is reduced in statistical estimation.
- 9) Statistics is a tool of comparison, investigation of data independent of type of filed.
- 10) Statistics is important in business, economics, industry, technology science and many more.

Limitations of Statistics

- ❏ Statistics does not Study Individuals.
- ❏ Statistics does not Study Qualitative Phenomena.
- ❏ Statistical Laws are not Exact
- ❏ Statistical Results are True only on an Average
- ❏ Statistical Relations do not Necessarily Bring Out the ‘Cause and Effect’

Relationship between Phenomena

Important Terms

- **Population**

The collection of **all** responses, measurements, or counts that are of interest.

- **Sample**

A portion or subset of the population.



- Parameter:

A number that describes a population characteristic.

*Average gross income of all people
in the United States in 2002.*

- Statistic:

A number that describes a sample characteristic.

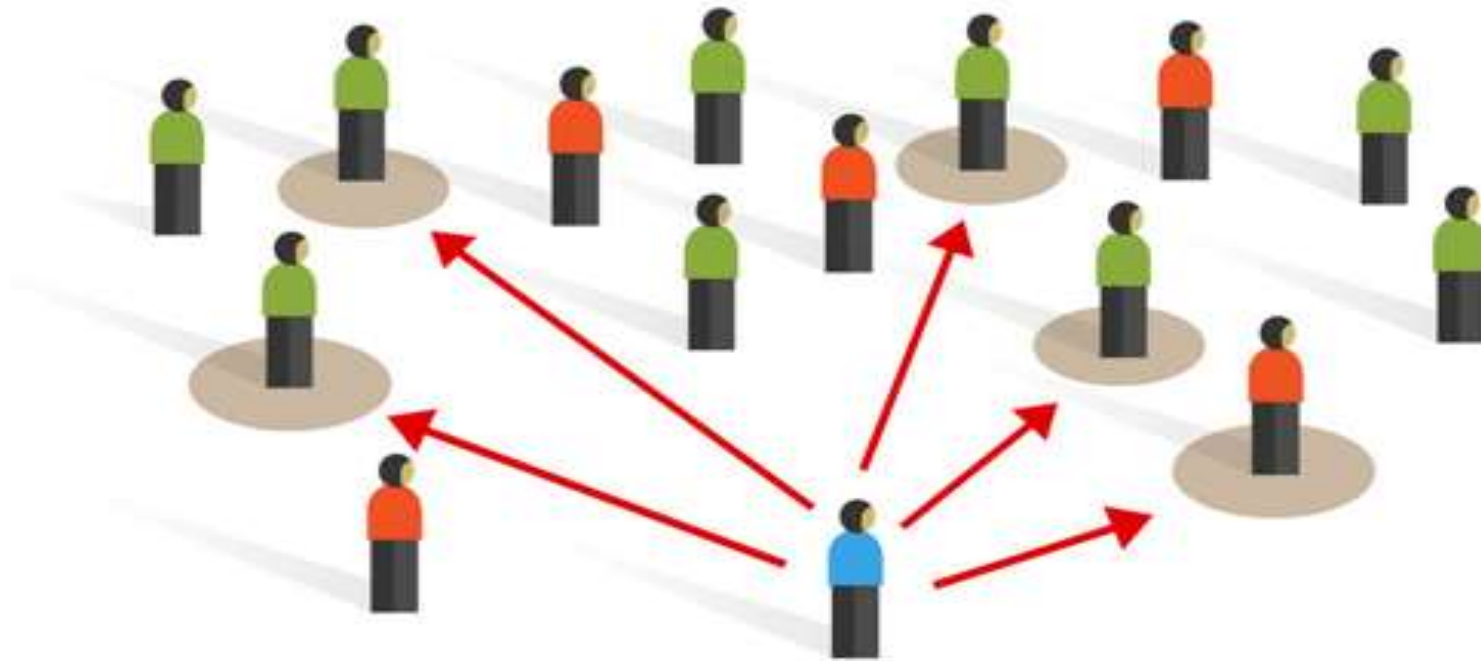
2002 gross income of people from a sample of three states.

Types of sampling

- 1 Purposive sampling
- 2 Random sampling
- 3 Simple sampling
- 4 Stratified sampling

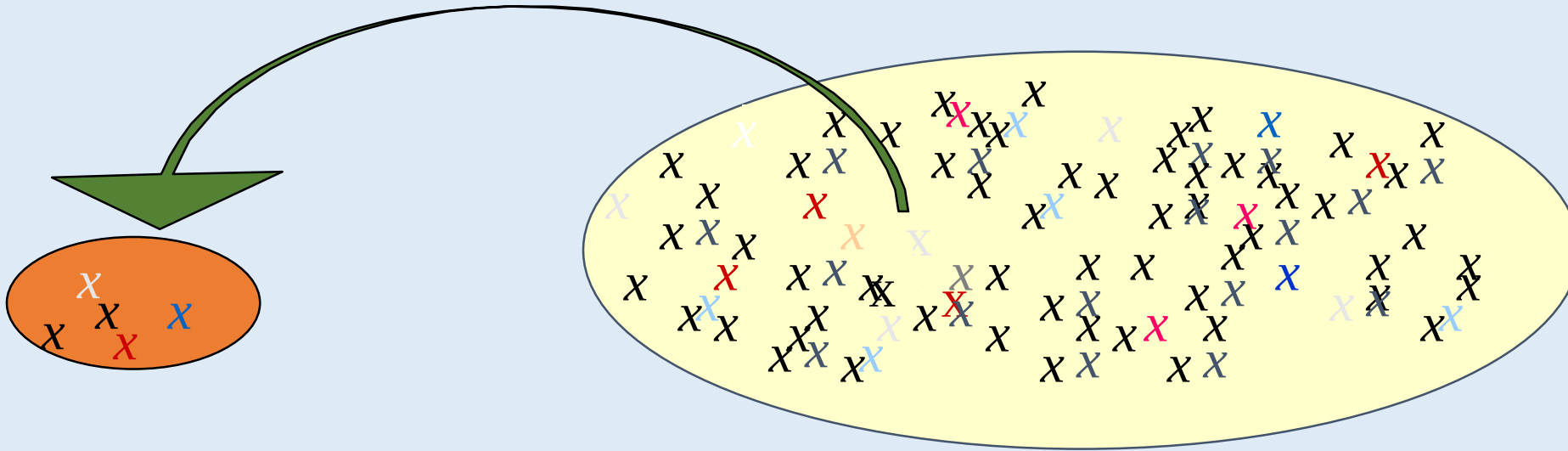
Purposive Sampling

- ❑ The sample units are selected with definite purpose in view.
- ❑ It is less costly and less time consuming.



Random Sample: Each member of the population has an equal chance of being selected.

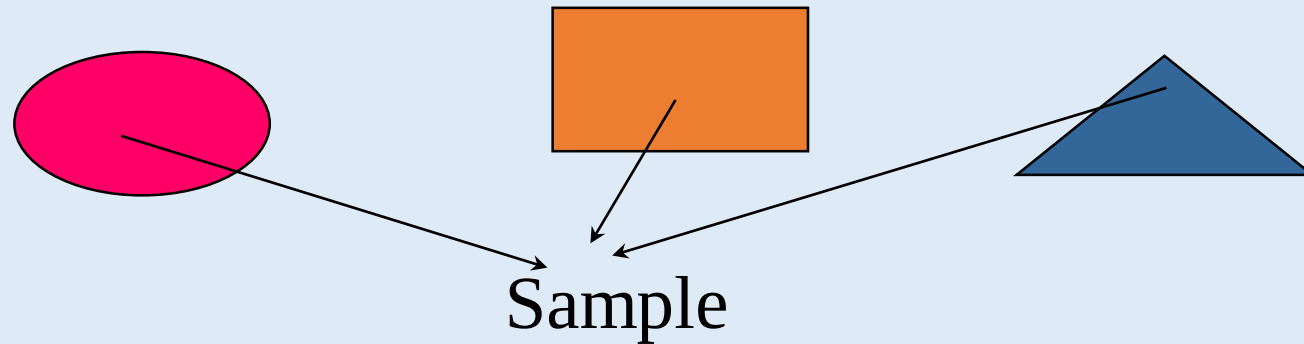
Simple Random Sample: All samples of the same size are equally likely.



- ✓ Assign a number to each member of the population.
- ✓ **Random numbers** can be generated by a random number table, software program or a calculator.
- ✓ Data from members of the population that correspond to these numbers become members of the sample.

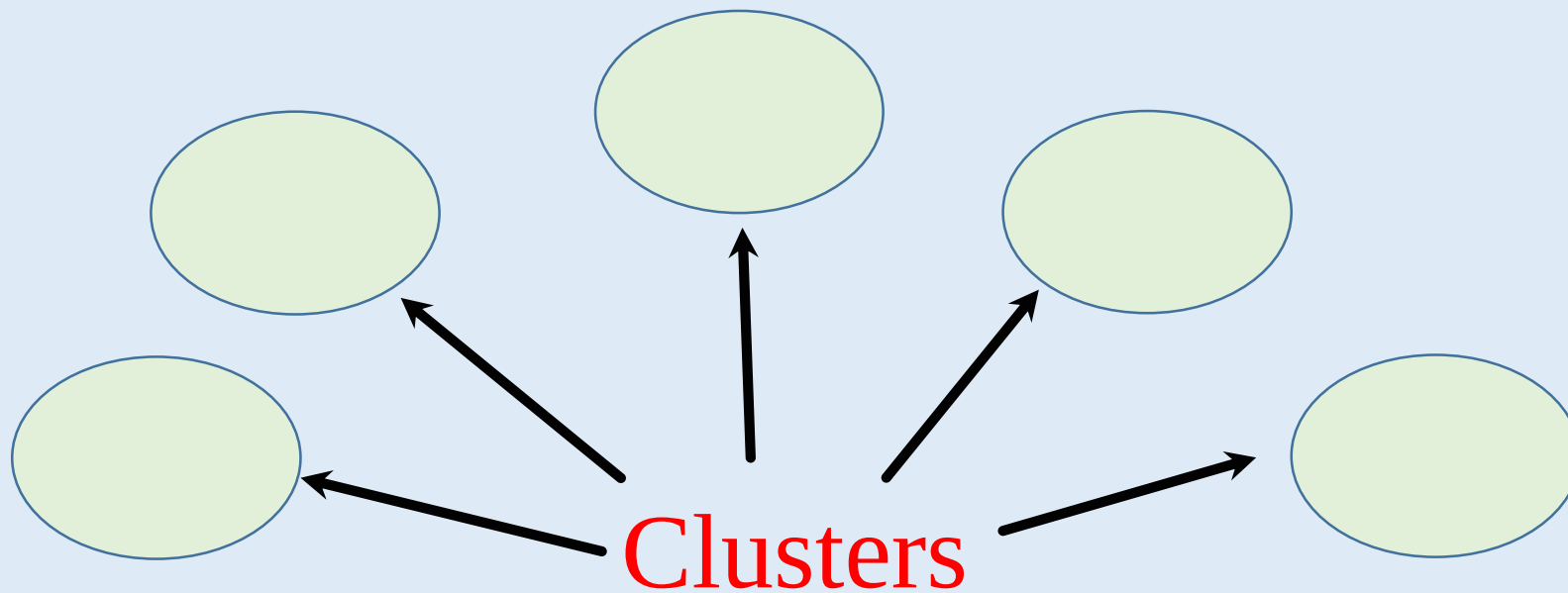
Stratified Random Samples

Divide the population into groups (strata) and select a random sample from each group. Strata could be age groups, genders or levels of education



Cluster Samples

Divide the population into individual units or groups and randomly select one or more units. The sample consists of all members from selected unit(s).



Systematic Samples

Choose a starting value at random. Then choose sample members at regular intervals.

X X

X X

X X

Sampling with and without Replacement

Population contains two types of sampling units.

1. A finite number of sampling unit.
2. An infinite number of sampling units.

Sampling can be done by **with replacement or without replacement**.

Sampling with replacement : When the same unit of the population involves in each sample more than once

Sampling without replacement: same unit of the population may not be included in each sample more than once.

If the size of population is N and sample size in n then

- If sampling is done with replacement, the total number of possible samples will amount to N^n .
- If sampling is done without replacement the total number of possible samples will be ${}^N C_n$

